ORIENTATION-BIT GEOMETRY OF THE MÖBIUS PATH: REFLECTION FIBRES, A DIVISOR-DEFECT GRAM FORM, AND THE RECOVERY OF SINE-KERNEL CONTENT FROM MEASURED QUANTITIES

Abstract. This notebook develops a single geometric object, the Gram form of the divisor defect on the orientation bits of the Möbius path, and uses it to separate what the quadratic profile of that path can and cannot see.

Three statements are proved here. First, the fibre of the quadratic profile $n \mapsto M(n)^2$ is exactly an orbit of an elementary abelian reflection group generated by excursion reflections. Second, the induced null on excursion signs is exactly isotropic, so that the expected energy of every eigenspace equals its dimension with no correction term. Third, and centrally, every cross-excursion contribution to the quadratic energy vanishes identically,

$$\sum_{a\in E_j}\sum_{b\in E_k}\mu(a)\mu(b)\,(N+1-\max(a,b))=0 \qquad (j\neq k),$$

because the increments of a complete excursion sum to zero. It follows that no statistic factoring through the quadratic profile can detect cross-excursion correlation of μ. The Gram form on orientation bits is introduced precisely because it is reflection-odd and therefore lives in the complement of that blind spot.

The notebook also isolates a methodological point. A statistic exhibits sine-kernel behaviour non-tautologically only when its value under the null is a number different from its value under the sine kernel; statistics whose null already equals the sine-kernel value, such as the mean square of $M(x)/\sqrt{x}$ and the energy deficit derived from it, carry no information about the zeros. Three measured quantities from separate work do satisfy the criterion, and for each the sine-kernel coefficient is written in closed form as a function of the measurement: through the consecutive spacing ratio and the superposition family $Y_2^{(p)}(x)=p^{-1}Y_2(x/p)$; through the moments $(n+1)^{-1}(n+2)^{-1}$ of the triangular form-factor deviation, which determine the sine kernel uniquely as a Hausdorff moment problem; and through the spacing variance of the lobe sequence.

A list of results established in separate work and used here as inputs is given in Section 2, with the status of each stated explicitly. Conjectures, heuristics and negative results are labelled throughout. No numerical computation is performed.

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Remark 0.1. This notebook is exploratory in nature.

Whenever a statement is rigorous it is stated inside a theorem, lemma, proposition or corollary environment and a proof is supplied. Results established in separate work are collected in Section 2 and are stated without proof, marked as such. All remaining claims are heuristic or conjectural and are labelled accordingly.

No numerical computation is performed anywhere in this notebook. The empirical quantities appearing in Section 8 are inputs recorded in separate work, not results obtained here.

Warning 0.2. Bibliographic metadata in Section 12 has not been verified against primary sources in this version. DOIs are omitted wherever they were not confirmed. Entries should be checked before this notebook is used as a source.

1. Introduction

1.1. **The object.** Let μ denote the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n) \tag{1.1}$$

the Mertens function, regarded as a ± 1 walk supported on the squarefree integers. We study this walk through three successive conditionings: on the squarefree mask, on the number of signs of each type, and on the quadratic profile $n \mapsto M(n)^2$.

The third conditioning is the subject of the notebook. We prove in Section 4 that the fibre of the quadratic profile is exactly an orbit of an elementary abelian reflection group generated by excursion reflections, and that the uniform measure on that group is a null model with no free parameter whatsoever. We then carry a symmetric bilinear form built from the divisor defect down to the quotient, obtaining a form on the orientation bits of the excursion decomposition.

1.2. **What is proved here.** Three results are proved in this notebook.

(i) *Reflection fibre* (Proposition 4.4). The fibre of the quadratic profile through a sign vector ε is exactly the orbit $\mathcal{R}_N \cdot \varepsilon$ of the excursion reflection group $\mathcal{R}_N \cong (\mathbb{Z}/2)^{Z(N)}$, where $Z(N)$ is the number of return points of M below N. Both inclusions are proved.

(ii) *Exact isotropy* (Proposition 4.6). Under the uniform measure ν on $\mathcal{R}_N$ the excursion signs satisfy $\mathbb{E}_\nu[ss^{\mathsf{T}}] = I_Z$ exactly, so that $\mathbb{E}_\nu\langle s, \widehat{P}s\rangle = \operatorname{rank}\widehat{P}$ with no correction term, and the variance is given in closed form.

(iii) *Blindness of the quadratic profile* (Theorem 6.2 and Corollary 6.3). For $j \neq k$,

$$\sum_{a\in E_j}\sum_{b\in E_k}\mu(a)\mu(b)\,(N+1-\max(a,b))=0, \tag{1.2}$$

because the sum of the increments over a complete excursion vanishes. Consequently the quadratic energy depends only on within-excursion two-point data, and no statistic factoring through the quadratic profile can detect cross-excursion correlation of μ.

Statement (iii) is the reason the Gram form on orientation bits is introduced at all: by Proposition 5.4 that form is reflection-odd, hence lives exactly in the complement of the blind spot identified by (1.2).

1.3. **What is taken as input.** The notebook draws on a body of results established in separate work. They are listed in full, with status, in Section 2; two of them are used actively and the remainder provide context. For orientation, the actively used ones are:

- the tensor factorisation $\widehat{K}(r,q)=\prod_p L_p(v_p(r),v_p(q))$ over all of $\mathbb{N}\times\mathbb{N}$, with explicit tridiagonal local factors, which enters as hypothesis (G2) of Definition 5.2 and is what makes the eigenbasis of the Gram form a character basis of the multiplicative structure;
- the exact permutation-null mean $\mathbb{E}_\pi[E_N]/N = 1/\pi^2 + M(N)^2/(3N) + o(1)$, which is the second rung of the conditioning hierarchy of Section 3.

Four further ingredients, referred to as **(N1)**–**(N4)**, supply the null model, its error bars, the central construction, and an interface to Robin-type criteria; they are introduced where they are used.

1.4. **The methodological point.** Section 8 isolates a criterion that we regard as the main conceptual content of Version 2.0.

The sine kernel is the kernel of a flat band projector (Lemma 8.1). Consequently any framework whose null model is a flat band reproduces the sine kernel automatically, and the appearance of sine-kernel behaviour in such a framework is a restatement of the null rather than evidence about the zeros of ζ. This applies in particular to the mean square of $M(x)/\sqrt{x}$ in logarithmic average and to the energy deficit derived from it; by Corollary 6.3 both are reflection-invariant and therefore contain no cross-excursion two-point mass at all (Remark 8.4).

The criterion that separates the informative case is the following.

> A statistic exhibits sine-kernel content non-tautologically if and only if its value under the null is a number different from its value under the sine kernel. The measured position between the two numbers is then the sine-kernel coefficient.

Three quantities measured in separate work satisfy this criterion, and for each of them the sine kernel is written in Section 8 as an explicit function of the measurement: the consecutive spacing ratio through the superposition family (8.2), the freeness-defect sequence through the triangular moments (8.5), and the lobe spacing variance through the anchors (8.7).

2. Standing results used as inputs

The following results are established in separate work by the author. They are stated here without proof and are used, where indicated, as inputs. No claim of novelty is made for them in this notebook, and no argument below reproves them.

Remark 2.1. *Unconditional results.*

(S1) Two-sided growth bounds for the antisymmetrised operator,

$$c\sqrt{N} \;\le\; \lambda_{\max}(D_N) \;\le\; C_\varepsilon\, N^{1/2+\varepsilon}, \tag{2.1}$$

together with the associated Cauchy interlacing corollaries. *Used as context in Remark 11.5.*

(S2) Trace-norm convergence of G_N/N^2 to a compact limit operator. *Context.*

(S3) The exact Chinese-remainder formula for the entries of Q_N. *Context; it is the source of (S4).*

(S4) The tensor factorisation

$$\widehat{K}(r,q) \;=\; \prod_{p} L_p(v_p(r), v_p(q)) \tag{2.2}$$

over all of $\mathbb{N}\times\mathbb{N}$, with explicit tridiagonal local factors L_p, and with the sign error in the corresponding proposition corrected. *Used actively: this is hypothesis (G2) of Definition 5.2.*

(S5) Positivity of the Hermitian collision coefficient: for every nontrivial zero $\rho=\beta+i\gamma$,

$$C_{\mathrm{herm}}(\rho) = \frac{2\,(\gamma^2-\beta^2)}{|\rho|^4\,|\zeta'(\rho)|^2} \;>\; 0, \tag{2.3}$$

established without assuming the Riemann hypothesis. *Context; see Remark 2.2.*

(S6) The closed form of the second-order curvature of the twisted product at $\vartheta=0$: for squarefree n with r prime factors,

$$z_0''(n) = -\frac{r^2}{2}\,\mathrm{Var}_{p\mid n}(\log p)\,, \tag{2.4}$$

together with its Dirichlet series

$$K(s) = -\sum_n \frac{z_0''(n)}{n^s} = \frac{\zeta(s)}{2}\left(S_0S_2 - S_1^2\right), \qquad S_2 = \left(\zeta'/\zeta\right)'. \tag{2.5}$$

Context.

(S7) The closed form of $z_\vartheta(n)$ for squarefree n with r prime factors, as a finite cosine sum over ordered partitions of the prime set via the cocycle $\Phi(\pi)$, with exact fourth-order coefficients

$$A_r = \frac{3(6r-11)}{8}, \qquad B_r = \frac{r(6r-11)}{8}, \tag{2.6}$$

valid for every r. *Context; it is the source of the twisted action considered in Conjecture 8.21.*

(S8) Blindness to Θ of the entire tower of fixed transversal jets, with complete proof, via the critical root of $P_x(z)=\sum_n \mu(n)^2 z^{\Omega(n)}$ near $z=-1$, the limit $\Phi_x(u)\to ue^{-u}$ uniform on compacta, and Cauchy bounds. *Context; the restriction to* fixed *jets is what leaves the scaling window of Conjecture 8.21 open.*

(S9) The unconditional equivalence between the Riemann hypothesis and tail stability of a two-channel positive logarithmic deformation built from the ordinal-prime transform. *Context; see Question 10.1.*

(S10) The exact permutation-null mean

$$\frac{\mathbb{E}_\pi[E_N]}{N} = \frac{1}{\pi^2} + \frac{M(N)^2}{3N} + o(1), \qquad E_N = \frac{1}{N}\sum_{n\le N} M(n)^2. \tag{2.7}$$

Used actively in Section 3.

Remark 2.2. One consequence of (S5) deserves recording, since it narrows a gap that was previously stated more broadly. At the extremal abscissa $s=2\Theta$ every contribution arises from zeros with $\beta=\Theta$ and carries the coefficient (2.3), which is positive. A sum of positive terms does not cancel, so cancellation between distinct pairs cannot occur *at the extremal abscissa.* What remains open is the case in which Θ is a supremum that is not attained, where a limiting or Tauberian formulation is required. We do not treat it here.

3. The three-constraint model

3.1. Mask, bits, and the fixed-count bridge.

Definition 3.1. Fix $N \geq 2$. The *squarefree mask* at level N is

$$\mathcal{M}_N = \{\, n \leq N : \mu(n) \neq 0 \,\}, \qquad m = m(N) = |\mathcal{M}_N|, \tag{3.1}$$

and the *orientation bits* of the Möbius path are

$$\varepsilon = (\varepsilon_n)_{n \in \mathcal{M}_N} \in \{\pm 1\}^m, \qquad \varepsilon_n = \mu(n). \tag{3.2}$$

Definition 3.2. Let $m_\pm$ be the number of $n \in \mathcal{M}_N$ with $\mu(n) = \pm 1$, so $m_+ + m_- = m$ and $m_+ - m_- = M(N)$. The *fixed-count sign bridge* π is the uniform measure on the $\binom{m}{m_+}$ vectors of $\{\pm 1\}^m$ with m_+ entries equal to $+1$. Write

$$\bar{\varepsilon} = \frac{M(N)}{m}. \tag{3.3}$$

Definition 3.2 is ingredient **(N2)**: mask frozen, sign count frozen, only the arrangement randomised.

3.2. Exact second moment of the bridge.

Proposition 3.3. *Under π,*

$$\mathbb{E}_\pi\Big[\varepsilon\varepsilon^{\mathsf{T}}\Big] = (1-\beta)\, I + \beta\, J, \qquad \beta = \frac{m\bar{\varepsilon}^{\,2} - 1}{m-1}, \tag{3.4}$$

with I the identity and J the all-ones matrix of size m.

Proof. The diagonal entries are $\varepsilon_n^2 = 1$. For $n \neq n'$, counting ordered pairs of distinct positions,

$$\mathbb{E}_\pi[\varepsilon_n \varepsilon_{n'}] = \frac{m_+(m_+-1) + m_-(m_- - 1) - 2m_+m_-}{m(m-1)} = \frac{(m_+ - m_-)^2 - m}{m(m-1)}. \tag{3.5}$$

Since $m_+ - m_- = m\bar{\varepsilon}$, the numerator is $m(m\bar{\varepsilon}^{\,2} - 1)$. □

Corollary 3.4. *For an orthogonal projection P on $\mathbb{R}^m$ of rank d,*

$$\mathbb{E}_\pi[\langle \varepsilon, P\varepsilon\rangle] = (1-\beta)\, d + \beta\, \|P\mathbf{1}\|^2, \tag{3.6}$$

and if $M(N) = o(\sqrt{m})$ then $\beta = O(1/m)$ and

$$\mathbb{E}_\pi[\langle \varepsilon, P\varepsilon\rangle] = d + O(1). \tag{3.7}$$

Proof. Trace (3.4) against P, using $\operatorname{tr} P = d$ and $\operatorname{tr}(PJ) = \|P\mathbf{1}\|^2$. For the second statement, $|\beta| \leq 1/(m-1)$ once $M(N)^2 \leq m$, and $\|P\mathbf{1}\|^2 \leq m$. □

Remark 3.5. Ingredient **(N1)**, the factorisation of $\operatorname{Cov}_\pi(M(n)^2, M(n')^2)$ through the three-block decomposition determined by $(n,\, n'-n,\, N-n')$ and the resulting $O(m)$ evaluation of the variance, is recorded in separate work and is not reproduced here. Only its structural consequence is used, namely that the covariance kernel is separable of bounded rank, so that the variance of any linear statistic of the prefix squares is a finite combination of power sums of the mask counting function. By Proposition 4.6 the third conditioning replaces this by an exact closed form, so **(N1)** is needed only at the level of the bridge.

4. The reflection fibre of the quadratic profile

This section and the next develop ingredient **(N3)**.

Definition 4.1. The *quadratic profile* at level N is

$$\Pi_N(\varepsilon) = \Big(M(n)^2\Big)_{n\le N}. \tag{4.1}$$

Definition 4.2. Let $0 = z_0 < z_1 < \cdots < z_Z$ enumerate $\{n \le N : M(n) = 0\} \cup \{0\}$, with $Z = Z(N)$. The *excursions* are

$$E_j = \{\, n : z_{j-1} < n \le z_j \,\}, \qquad j = 1, \dots, Z, \tag{4.2}$$

together with a terminal block $E_{Z+1} = \{n : z_Z < n \le N\}$, empty exactly when $M(N) = 0$.

Definition 4.3. The *excursion reflection group* is $\mathcal{R}_N \cong (\mathbb{Z}/2)^{Z(N)}$, acting by

$$(\sigma \cdot \varepsilon)_n = \sigma_{j(n)}\, \varepsilon_n, \qquad \sigma \in \{\pm 1\}^Z, \tag{4.3}$$

where $j(n)$ indexes the excursion containing n and the terminal block is held fixed.

Proposition 4.4. *For every $\sigma \in \mathcal{R}_N$ and every $n \le N$,*

$$M_{\sigma\cdot\varepsilon}(n) = \sigma_{j(n)}\, M_\varepsilon(n), \tag{4.4}$$

so Π_N is $\mathcal{R}_N$-invariant. Conversely, two sign vectors on the same mask with the same quadratic profile differ by an element of $\mathcal{R}_N$. Hence the fibre of Π_N through ε is exactly the orbit $\mathcal{R}_N \cdot \varepsilon$.

Proof. For $n \in E_j$ we have $M_\varepsilon(z_{j-1}) = 0$, hence $M_\varepsilon(n) = \sum_{z_{j-1}<k\le n} \varepsilon_k$ is a sum entirely inside E_j; applying σ multiplies every summand by σ_j, giving (4.4).

Conversely, let $\varepsilon, \varepsilon'$ have the same mask and satisfy $M_\varepsilon(n)^2 = M_{\varepsilon'}(n)^2$ for all n. The two paths then have the same zero set, hence the same excursion decomposition, and $|M_\varepsilon| = |M_{\varepsilon'}|$ pointwise. On each E_j the path does not vanish in the interior, so M_ε has constant sign there; let σ_j be the ratio of the two signs. A path on a fixed mask is determined by its increments, and the increments of $M_{\varepsilon'}$ on E_j are σ_j times those of M_ε, so $\varepsilon' = \sigma \cdot \varepsilon$. □

4.1. The canonical null on orientation bits.

Definition 4.5. The *excursion signs* of ε are $s = (s_1, \dots, s_Z) \in \{\pm 1\}^Z$, where s_j is the sign of M_ε in the interior of E_j. The *reflection null* ν is the uniform measure on $\{\pm 1\}^Z$.

Proposition 4.6. *Under ν,*

$$\mathbb{E}_\nu\Big[ss^{\mathsf{T}}\Big] = I_Z \tag{4.5}$$

exactly, with no rank-one correction. Hence for symmetric A,

$$\mathbb{E}_\nu[\langle s, As\rangle] = \operatorname{tr} A, \qquad \operatorname{Var}_\nu[\langle s, As\rangle] = 2\sum_{j\ne k} A_{jk}^2, \tag{4.6}$$

and for an orthogonal projection $\widehat{P}$ of rank d,

$$\mathbb{E}_\nu\Big[\langle s, \widehat{P}s\rangle\Big] = d, \qquad \operatorname{Var}_\nu\Big[\langle s, \widehat{P}s\rangle\Big] = 2\left(d - \sum_j \widehat{P}_{jj}^2\right). \tag{4.7}$$

Proof. The s_j are independent Rademacher variables under ν, giving (4.5). Writing $\langle s, As\rangle = \operatorname{tr} A + \sum_{j\ne k} A_{jk} s_j s_k$ and using $\mathbb{E}_\nu[s_j s_k s_{j'} s_{k'}] = 1$ exactly when $\{j,k\} = \{j',k'\}$ gives the variance. For a projection, $\sum_{j,k} \widehat{P}_{jk}^2 = \operatorname{tr} \widehat{P}^2 = d$. □

Remark 4.7. Corollary 3.4 and Proposition 4.6 form a hierarchy. Conditioning on the mask removes the squarefree density, the source of the term $1/\pi^2$ in (2.7); conditioning on the sign count removes the endpoint drift $M(N)^2/(3N)$; conditioning in addition on the quadratic profile leaves exactly $\mathcal{R}_N$, whose null is isotropic *without* the $O(1)$ correction of (3.7). The third conditioning is the step at which the null becomes parameter-free.

5. The divisor-defect Gram form on orientation bits

Definition 5.1. Let B be the divisor-defect matrix attached to $\mathcal{M}_N$ and

$$G = B^{\mathsf{T}} B \tag{5.1}$$

its Gram form on $\mathbb{R}^m$.

Definition 5.2. G is *admissible* if:

(G1) G is real symmetric and positive semidefinite;
(G2) the associated kernel admits the tensor factorisation (2.2) of (S4), with local tridiagonal factors indexed by prime valuations.

Property (G2) is (S4) and is not proved here. Its role is to make the eigenbasis of G a character basis of the multiplicative structure; see Remark 8.2.

Definition 5.3. The *orientation-bit form* is

$$\widehat{G}_{jk} \;=\; \sum_{a\in E_j}\sum_{b\in E_k} G_{ab}, \qquad 1 \le j,k \le Z, \tag{5.2}$$

with spectral resolution

$$\widehat{G} = \sum_k \lambda_k \widehat{P}_k, \qquad d_k = \operatorname{rank} \widehat{P}_k. \tag{5.3}$$

Proposition 5.4. *The form $s \mapsto \langle s, \widehat{G}s\rangle$ is invariant under the global flip $s \mapsto -s$, and its off-diagonal part is odd under every individual reflection $s_j \mapsto -s_j$. In particular $\langle s, \widehat{G}s\rangle$ is not a function of the quadratic profile unless $\widehat{G}$ is diagonal.*

Proof. The term $s_j s_k$ changes sign under $s_j \mapsto -s_j$ for $j \neq k$, while $\widehat{G}_{jj} s_j^2 = \widehat{G}_{jj}$ is constant. By Proposition 4.4 any function of the profile is $\mathcal{R}_N$-invariant. □

6. Blindness of the quadratic profile

Lemma 6.1. *For every $N \ge 1$,*

$$\sum_{n\le N} M(n)^2 = \sum_{a,b\le N} \mu(a)\mu(b)\,(N+1-\max(a,b))\,. \tag{6.1}$$

Proof. Expand $M(n)^2 = \sum_{a,b\le n}\mu(a)\mu(b)$ and exchange the order of summation; the pair (a,b) occurs for exactly $N+1-\max(a,b)$ values of n. □

Theorem 6.2. *Assume $M(N) = 0$, so the excursion decomposition has no terminal block. Then for all $j \neq k$,*

$$\sum_{a\in E_j}\sum_{b\in E_k} \mu(a)\mu(b)\,(N+1-\max(a,b)) = 0. \tag{6.2}$$

Consequently $\sum_{n\le N} M(n)^2$ depends only on the within-excursion two-point data.

Proof. Let $j < k$, so every $a \in E_j$ precedes every $b \in E_k$ and $\max(a,b) = b$. The double sum factorises as

$$\left(\sum_{a\in E_j} \mu(a)\right)\left(\sum_{b\in E_k} \mu(b)\,(N+1-b)\right), \tag{6.3}$$

and $\sum_{a\in E_j} \mu(a) = M(z_j) - M(z_{j-1}) = 0$ by Definition 4.2. The case $j > k$ is symmetric. □

Corollary 6.3. *No statistic factoring through Π_N detects cross-excursion correlation of μ. In particular the energy, all of its moments, and all functionals of excursion heights and lengths are blind to it.*

Proof. A statistic factoring through Π_N is $\mathcal{R}_N$-invariant by Proposition 4.4; by Theorem 6.2 the cross-excursion two-point mass contributes nothing to the energy, and by (4.3) it is exactly the part of the two-point data on which $\mathcal{R}_N$ acts nontrivially. □

Remark 6.4. The vanishing in (6.2) is caused by the defining property of an excursion, that its increments sum to zero, and therefore cannot be repaired by reweighting: any weight depending on a and b only through $\max(a,b)$ yields the same factorisation (6.3).

Remark 6.5. If $M(N) \neq 0$ the terminal block is nonempty and (6.2) fails for pairs involving it, since $\sum_{a\in E_{Z+1}} \mu(a) = M(N)$; the correction is a single rank-one term proportional to $M(N)$. We normalise by taking N at a return point.

7. The scale-resolved statistic

Definition 7.1. With (5.3) and s the excursion signs of the true Möbius path,

$$\mathcal{E}_k = \langle s, \widehat{P}_k s\rangle, \qquad \mathcal{R}_k = \frac{\mathcal{E}_k}{d_k}. \tag{7.1}$$

Proposition 7.2. *Under ν,*

$$\mathbb{E}_\nu[\mathcal{R}_k] = 1 \quad \textit{exactly}, \qquad \mathrm{Var}_\nu[\mathcal{R}_k] = \frac{2}{d_k^2}\left(d_k - \sum_j \widehat{P}_{k,jj}^2\right), \tag{7.2}$$

and $\sum_k \mathcal{E}_k = \|s\|^2 = Z = \sum_k d_k$.

Proof. Apply Proposition 4.6 with $A = \widehat{P}_k$ and use $\sum_k \widehat{P}_k = I_Z$. □

Definition 7.3. The *Gram anisotropy* at level N is $k \mapsto \mathcal{R}_k - 1$.

Remark 7.4. The calibration (7.2) has no free parameter, no fitted constant and no asymptotic normalisation. By Proposition 5.4 and Corollary 6.3, $\mathcal{R}_k$ is reflection-odd and therefore measures exactly the information the quadratic profile cannot carry.

Conjecture 7.5. *The Gram anisotropy is not identically $o(1)$: there are eigenspaces of $\widehat{G}$ on which $\mathcal{R}_k - 1$ exceeds the null standard deviation (7.2) by an unbounded factor as $N \to \infty$.*

8. Sine-kernel content from measured quantities

8.1. Flat bands, and why they make the question empty.

Lemma 8.1. *Let P be the orthogonal projection onto a contiguous block of d characters of a cyclic group of order Z. In the position basis the kernel of P is a Dirichlet kernel, and under the scaling $Z \to \infty$ with d/Z fixed and positions rescaled by the mean spacing it converges to $\sin(\pi x)/(\pi x)$.*

Proof. $P(n,n') = Z^{-1}\sum_{t\in S} e^{2\pi i t(n-n')/Z}$ with S an interval of length d is a Dirichlet kernel, whose scaling limit is as stated; see [8]. □

Equivalently, on the Fourier side the unitary two-level form factor $\min(|\alpha|, 1)$ is the indicator of a band, and

$$1 - R_2(x) \;=\; Y_2(x) \;=\; \left(\frac{\sin \pi x}{\pi x}\right)^2 \;=\; \int_{-1}^{1} (1 - |\tau|)\, e^{2\pi i \tau x}\, d\tau, \tag{8.1}$$

the Fejér identity: the sine-kernel cluster function is the transform of the triangle. Montgomery's result gives the pair correlation for $|\alpha| < 1$ [9]; the plateau $|\alpha| > 1$ is open.

Remark 8.2. Hypothesis (G2) of Definition 5.2, that is (S4), states that the eigenbasis of G factorises over primes, hence is a character basis of the multiplicative structure rather than of an additive cyclic group. The band of Lemma 8.1, transported to positions, therefore produces a sine kernel in the variable $\log n$ rather than in n. That is the variable in which the explicit formula oscillates, since $n^{i\gamma} = e^{i\gamma \log n}$, so the identification is the natural one. It does not, however, affect Warning 8.3.

Warning 8.3. By Proposition 4.6 the reflection null has $\mathbb{E}_\nu[ss^{\mathsf{T}}] = I$, which restricted to a contiguous block of characters is, by Lemma 8.1, a sine kernel. Hence the sine kernel is the *null hypothesis* of $\mathcal{R}_k$, not a consequence of it. Deriving sine-kernel behaviour from flatness of the null is deriving the null from the null and carries no evidential weight concerning the zeros of ζ.

Remark 8.4. The same exclusion applies to two quantities that might appear to be candidates. The logarithmically averaged mean square of $M(x)/\sqrt{x}$ tends to $\sum_\rho |\rho\zeta'(\rho)|^{-2}$, a diagonal quantity; and the ratio of the true quadratic energy to (2.7) is a function of that same quantity. Both are $\mathcal{R}_N$-invariant, so by Corollary 6.3 neither contains any cross-excursion two-point mass. Neither admits a nontrivial sine-kernel reading, since neither sees Y_2 away from the origin.

8.2. **The criterion.**

Remark 8.5. A statistic carries sine-kernel content non-tautologically if and only if its value under the null is a number different from its value under the sine kernel. The measured position between the two numbers is then the sine-kernel coefficient. Warning 8.3 describes exactly the failure of this criterion; the three subsections below give quantities that satisfy it.

8.3. **Through the consecutive spacing ratio.**

Definition 8.6. For $p \geq 1$, the *superposition family* consists of the superposition of p independent identically distributed spectra, each unfolded to density $1/p$. Its two-level cluster function is

$$Y_2^{(p)}(x) = \frac{1}{p}\, Y_2\!\left(\frac{x}{p}\right), \tag{8.2}$$

so that with $Y_2 = (\sin \pi x/\pi x)^2$,

$$R_2^{(p)}(x) = 1 - \frac{1}{p}\left(\frac{\sin(\pi x/p)}{\pi x/p}\right)^2, \qquad Y_{jk}^{(p)} = \frac{1}{p}\left(\frac{\sin(\pi(j-k)/p)}{\pi(j-k)/p}\right)^2. \tag{8.3}$$

The coefficient $1/p$ is the *sine-kernel fraction*; $p = 1$ is the pure sine kernel and $p \to \infty$ is the vanishing of local repulsion.

Remark 8.7. The identity $\int Y_2^{(p)} = 1$ holds for every finite p, whereas a Poisson spectrum has $\int Y_2 = 0$; the limit $p \to \infty$ is therefore non-uniform between local behaviour and the sum rule. This does not affect the local reading used here.

Experiment 8.8. *Recorded in separate work, not performed here.* The consecutive spacing ratio statistic $\tilde{r}$ of the spectrum of the smooth arithmetic term $2N/[m, n]$ was measured and found to be approximately 0.42.

Remark 8.9. The reference values of $\tilde{r}$ are $2\ln 2 - 1 = 0.38629\ldots$ for a Poisson spectrum, approximately 0.5307 for the orthogonal class and approximately 0.5996 for the unitary class [1]; the last two are recalled and have not been verified here.

Proposition 8.10. *Suppose $\tilde{r}$ is strictly monotone in p along the family of Definition 8.6, with $\tilde{r}(1)$ the unitary value and $\tilde{r}(\infty) = 2\ln 2 - 1$. If a measured value satisfies $\tilde{r} > 2\ln 2 - 1$ strictly, then the corresponding p is finite and the sine-kernel fraction $1/p$ in (8.3) is strictly positive.*

Proof. Immediate from strict monotonicity and the stated endpoint values. □

Remark 8.11. Proposition 8.10 is the point at which the criterion of Remark 8.5 is met. A flat null forces $\tilde{r}$ to the single value $2\ln 2 - 1$ and can produce nothing else, whereas the value recorded in Experiment 8.8 exceeds it. The sine-kernel coefficient is then not assumed but determined, by solving $\tilde{r}(p) = 0.42$ for p and substituting into (8.3). The monotonicity hypothesis of Proposition 8.10 is standard for superpositions but is stated as a hypothesis, not proved here.

8.4. **Through the moments of the triangular deviation.** By (8.1), sine-kernel behaviour is equivalent to the two-level form factor deviating from its flat value by the triangle,

$$K(\tau) - 1 = -\,(1-\tau)_+ . \tag{8.4}$$

Lemma 8.12. *The moments of the triangle are*

$$m_n = \int_0^1 (1-\tau)\,\tau^n\,d\tau = \frac{1}{n+1} - \frac{1}{n+2} = \frac{1}{(n+1)(n+2)}. \tag{8.5}$$

Since $(1-\tau)_+$ is supported on the compact interval $[0,1]$, it is uniquely determined by $\{m_n\}_{n\geq 0}$.

Proof. The integral is elementary; uniqueness is the determinacy of the Hausdorff moment problem on a compact interval. □

Remark 8.13. Lemma 8.12 changes the logical status of a defect sequence. A deviation in the variance of a linear statistic of order k is a moment of (8.4) against a known weight. A sequence of such defects therefore does not merely agree with the sine kernel: by determinacy it *reconstructs* it. Two defects do not determine the triangle, but their ratio already discriminates.

Experiment 8.14. *Recorded in separate work, not performed here.* In a test of asymptotic freeness of the arithmetic operator against the shift, the relative defect from the free value was approximately 12% at order two and approximately 32% at order three.

Remark 8.15. The three candidate growth patterns from order two to order three are distinct. The triangle moments (8.5) give $m_3/m_2 = 12/20 = 0.6$, so they *decrease*. The variance of linear statistics in the Chebyshev basis is diagonal with variance growing linearly in the order [5], which would give $3/2 = 1.5$; this is recalled and not verified here. The defects of Experiment 8.14 give $32/12 \approx 2.67$. The observed growth is faster than linear, whereas the triangle requires decay. This discrepancy is a measured quantity and is the natural candidate for the non-sine-kernel part. The ratio is invariant under a global rescaling of the normalisation of "relative defect", so the discrimination does not depend on that convention; the individual moments do, and cannot be read without it.

8.5. **Through the lobe spacing variance.**

Definition 8.16. Let s_n denote the normalised spacing associated with the n-th lobe, so that $s_n = 1$ characterises the ideal lobe. The *lobe curvature energy* in a window of ordinals is

$$\mathcal{E}_{\mathrm{lobe}}(n_0, L) = \frac{1}{L} \sum_{n_0 < n \leq n_0 + L} (s_n - 1)^2 . \tag{8.6}$$

Lemma 8.17. *Under the Wigner surmises, the variance of the normalised spacing is*

$$\mathrm{Var}_{\mathrm{Poisson}}(s) = 1, \qquad \mathrm{Var}_{\mathrm{GOE}}(s) = \frac{4}{\pi} - 1, \qquad \mathrm{Var}_{\mathrm{GUE}}(s) = \frac{3\pi}{8} - 1. \tag{8.7}$$

Proof. For the surmise densities $p(s) = \frac{\pi}{2} s e^{-\pi s^2/4}$ and $p(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, both normalised to $\langle s \rangle = 1$, the ratios of Gaussian moments give $\langle s^2 \rangle = 4/\pi$ and $\langle s^2 \rangle = 3\pi/8$ respectively; the Poisson case is the exponential law. □

Remark 8.18. The three values in (8.7) are distinct, and the null value is 1, so (8.6) satisfies the criterion of Remark 8.5. It is the only observable considered in this notebook that is measured directly on ordinates rather than on an arithmetic surrogate. The reading is coarser than that of Section 8.3, since the gap distribution is not a linear functional of Y_2; the surmises are approximations and not exact random matrix values.

8.6. **A derived measurement.**

Remark 8.19. Applying Section 8.3 separately within each divisor class yields the map

$$d \longmapsto \frac{1}{p(d)}, \qquad \tilde{r}(p(d)) = \tilde{r}_d, \tag{8.8}$$

the profile of sine-kernel fraction resolved by divisor class. This is a derived measurement rather than a new one: it separates the coefficient by arithmetic scale instead of aggregating it. The systematic observation recorded in separate work, that the anomaly at $d = 1$ is the weakest with stronger decorrelation for $d \geq 2$, predicts that $1/p(1)$ is the minimum of the profile. This prediction is falsifiable and requires no new theory.

8.7. **The symmetry obstruction and the twist.**

Remark 8.20. The group $\mathcal{R}_N \cong (\mathbb{Z}/2)^Z$ and the form $\widehat{G}$ are real. A real symmetric ensemble has $\beta = 1$ statistics, whose two-level form factor $b(\tau) = 2\tau - \tau \log(1 + 2\tau)$ on $\tau \leq 1$ [8] is not the indicator of a band and has no plateau; the orthogonal two-level function contains the sine kernel but is not $1 - (\sin \pi x/\pi x)^2$. The conjectural class for the zeros of ζ is $\beta = 2$ [9, 10]. The present construction is therefore in the wrong symmetry class, and the obstruction is structural.

Conjecture 8.21. *Let $U(1)^Z$ act on orientation bits by $s_j \mapsto e^{i\vartheta_j} s_j$, extending* (4.3) *from $(\mathbb{Z}/2)^Z \subset U(1)^Z$, and let $\widehat{G}^{(\vartheta)}$ be the Hermitian form obtained by transporting G through the twisted action, with ϑ governed by the cocycle of (S7). Then the local statistics of $\widehat{G}^{(\vartheta)}$ interpolate between the orthogonal and unitary classes, with crossover at $\vartheta_c \asymp 1/Z$, that is, when the antisymmetric part of the form is comparable to the mean level spacing.*

Remark 8.22. Conjecture 8.21 is modelled on the known interpolating ensembles [12]. What is conjectural is not the existence of such an interpolation but the claim that the twist of orientation bits realises it for $\widehat{G}$ at the stated scale. Note that (S8) closes the tower of *fixed* transversal jets, and therefore does not cover the scaling window $\vartheta \asymp 1/Z$.

Warning 8.23. Even if Conjecture 8.21 holds it would establish only that the ensemble lies in a symmetry class compatible with the zeros of ζ. That is necessary and very far from sufficient.

Question 8.24. Is there a formulation of the plateau $F(\alpha) = 1$ for $\alpha > 1$ as a flatness statement for $\widehat{G}^{(\vartheta)}$ on a band of eigenspaces, or equivalently as flatness of (8.6) across a window of ordinals? A positive answer would be a reformulation, not a proof, but it would move the plateau into a setting carrying a transfer structure along the excursion index.

9. Route toward two-point cancellation

Remark 9.1. Combining Lemma 6.1, Theorem 6.2 and Proposition 5.4, the two-point data of μ splits into a reflection-even part carried entirely by the quadratic profile and entirely within excursions, and a reflection-odd part consisting of all cross-excursion pairs. Shifted pairs $\mu(a)\mu(a + h)$ are cross-excursion for all but a vanishing proportion of a once h exceeds typical excursion lengths, so the content of two-point cancellation lies in the reflection-odd part, which is what $\mathcal{R}_k$ measures.

Conjecture 9.2. *With*

$$C_N(h) = \sum_{a \le N-h} \mu(a)\mu(a+h), \tag{9.1}$$

the aggregate cross-excursion mass of $\{C_N(h)\}_h$ *is, in the mean, of the order predicted by* ν*; equivalently the Gram anisotropy is bounded on the relevant eigenspaces.*

Heuristic 9.3. The triangular weight in (6.1) aggregates all shifts up to N with comparable mass. Logarithmic weighting is the regime in which two-point cancellation for multiplicative functions is presently accessible [14], building on [7]. This suggests replacing the triangular weight by a logarithmic one before pushing to orientation bits. The remaining obstruction is uniformity in h: the available theorems are for fixed or slowly growing h, while the aggregate requires h up to N. We do not claim this gap can be closed; on this route it is the only gap.

Question 9.4. Does passing to $\mathcal{R}_k$ change the uniformity requirement in Heuristic 9.3? Since $\widehat{G}$ aggregates pairs by excursion block rather than by shift, the relevant uniformity is in the block index, and the two need not be equivalent.

10. The Robin interface

Ingredient **(N4)** enters here. The form G is built from the divisor defect; Robin's criterion [13] is an inequality on $\sigma(n)$ equivalent to the Riemann hypothesis, with elementary reformulations [6].

Question 10.1. Suppose the truncated functional attached to G satisfies an exact elementary Robin-type equation. Does its positivity cone coincide with the cone associated with the tail-stability criterion (S9)? If the cones coincide the two criteria are reparametrisations of one another; if they differ the pair yields a positivity family not reducible to either.

Warning 10.2. The literature on Robin-type criteria is extensive. No claim of novelty is made for **(N4)**, and the exact elementary form should be checked against the literature before being described as new.

11. Negative results and failed routes

Remark 11.1. The scalar quadratic energy at a single level N carries no local information: its cross terms appear with a weight essentially constant on the scale of the mean spacing of ordinates. The defect is not the weight but the collapse to a single number; as a function of $\log N$ the energy is almost periodic with frequency set given by differences of ordinates and an invertible multiplier at every real frequency. The present framework subsumes this without the auxiliary variable.

Remark 11.2. Within the family of statistics built from M alone, the weights attached to individual zeros decay in the ordinate, so the resulting pair-correlation measure mixes heights. This cannot be repaired by any operation on $\log N$, which is conjugate to differences of ordinates and not to the ordinates themselves.

Remark 11.3. The excursion decomposition admits a classical passage to continued fractions and hence to a Jacobi matrix [3], which is real symmetric and therefore lands in the orthogonal class, as in Remark 8.20. The coincidence of the two obstructions, in the excursion picture and in the Gram picture, motivates Conjecture 8.21.

Remark 11.4. A cell-complex description of the Mertens function exists, with $M(n)$ an Euler characteristic and Betti numbers given by odd divisor sums [2]. It does not reach two-point cancellation, because the complex is multiplicative while shifted correlations are additive: there is no map relating the complex at n to that at $n+h$. A complex indexed by pairs $(n, n+h)$ with Euler characteristic $C_N(h)$ would remove the obstruction; we are not aware of such an object and make no claim that it exists.

Remark 11.5. Any operator whose spectrum is the set of ordinates must reproduce the counting function $N(T) \sim (T/2\pi)\log(T/2\pi)$. The bounds (2.1) of (S1) place $\lambda_{\max}(D_N)$ at the Wigner scale, whose integrated density is semicircular and fails that requirement by shape. D_N as it stands is therefore not a candidate spectral realisation.

Remark 11.6. Any hierarchy of bounds on the Möbius walk must respect the unconditional oscillation result $M(x) = \Omega_{\pm}(\sqrt{x})$ with explicit constants [11]; no ε in such a hierarchy is removable.

12. Open questions and future work

(Q1) Does $\widehat{G}$ of (5.2) inherit the tensor factorisation (2.2) from G? The excursion blocks are defined by the zero set of M and respect no multiplicative structure, so a clean inheritance is not expected; a partial inheritance restricted to the local factors L_p with small p would be the interesting outcome. This is a finite and elementary question, it requires no numerical work, and everything in Sections 7–9 depends on it. It is the pivotal technical question of the notebook.

(Q2) Establish or refute Conjecture 7.5, and determine which eigenspaces carry the anisotropy.

(Q3) Test the prediction of Remark 8.19 that $1/p(1)$ is the minimum of the divisor-resolved sine-kernel fraction.

(Q4) Fix the normalisation convention under which the defects of Experiment 8.14 are moments, so that Remark 8.15 can be read quantitatively and not only through the ratio.

(Q5) Determine the growth of $Z(N)$. Every statement in Sections 7–9 is uniform in Z, but the strength of the conclusions depends on it, and no claim is made here.

(Q6) Close the Tauberian case of Remark 2.2.

(Q7) Settle Question 8.24, Question 9.4 and Question 10.1, the last subject to Warning 10.2.

Notebook History

Version 1.0: Three-constraint model and exact bridge second moment; reflection fibre of the quadratic profile and the canonical parameter-free null; divisor-defect Gram form pushed to orientation bits; blindness theorem for the quadratic profile; scale-resolved statistic with exact mean and variance; exploratory routes toward sine-kernel statistics and two-point cancellation; negative results recorded.

Version 2.0: Abstract added. Introduction restructured to state separately what is proved here, what is taken as input, and the methodological point. Section 2 added, listing the results established in separate work with the status and role of each, together with the narrowing of the Hermitian-collision gap to the non-attained supremum case in Remark 2.2. Section 8 rewritten around the criterion of Remark 8.5 and extended with three routes recovering the sine-kernel coefficient from measured quantities: the superposition family and the spacing ratio, the triangular moments and Hausdorff determinacy, and the lobe spacing variance; the divisor-resolved sine-kernel fraction added as a derived measurement. Question (Q1) promoted to pivotal status.

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